

Hartree-Fock Method

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Outline

- Fock operator
- Self Consistent Field method
- Total energy (true for both ground state and excited states)
- Ground state and excited state
- Post-HF methods

Fock Operator

- Fock operator is one-electron operator.
- For N-electron system, it solves N one-electron Schrödinger equations using below equations:

$$\hat{F}(\mathbf{x}_k)\phi_k(\mathbf{x}_k) \equiv \left[\hat{h}(\mathbf{x}_k) + \hat{J}(\mathbf{x}_k) - \hat{K}(\mathbf{x}_k) \right] \phi_k(\mathbf{x}_k) = \epsilon_k \phi_k(\mathbf{x}_k),$$

where the **Coulomb operator** $\hat{J}(\mathbf{x}_k)$ and the **exchange operator** $\hat{K}(\mathbf{x}_k)$ are defined as follows

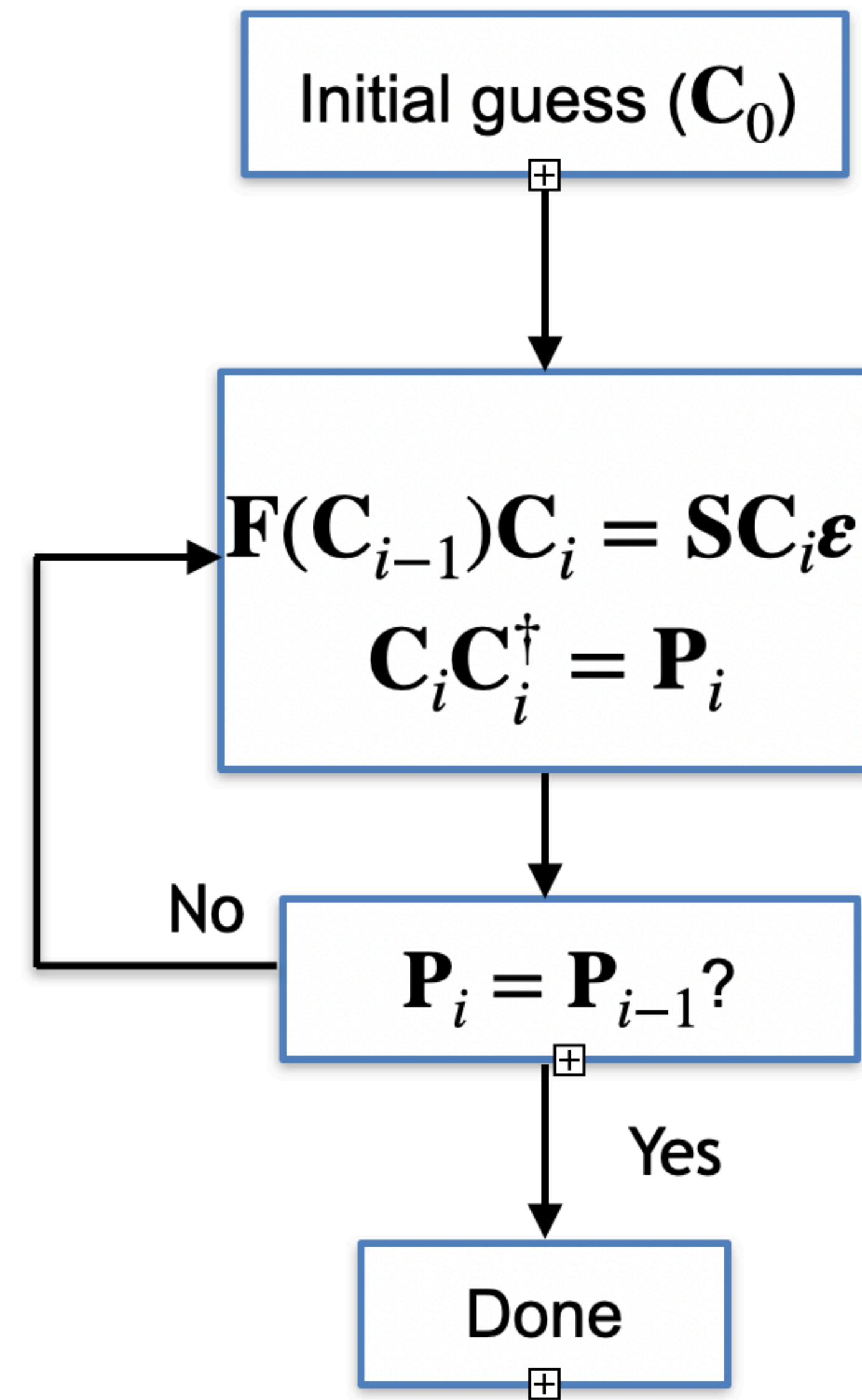
$$\hat{J}(\mathbf{x}_k) \equiv \sum_{j=1}^N \int d\mathbf{x}_j \frac{\phi_j^*(\mathbf{x}_j)\phi_j(\mathbf{x}_j)}{|\mathbf{x}_k - \mathbf{x}_j|} = \sum_{j=1}^N \int d\mathbf{x}_j \frac{\rho(\mathbf{x}_j)}{|\mathbf{x}_k - \mathbf{x}_j|},$$

$$\hat{K}(\mathbf{x}_k)\phi_k(\mathbf{x}_k) \equiv \sum_{j=1}^N \phi_j(\mathbf{x}_k) \int d\mathbf{x}_j \frac{\phi_j^*(\mathbf{x}_j)\phi_k(\mathbf{x}_j)}{|\mathbf{x}_k - \mathbf{x}_j|}.$$

- ϵ_k here is canonical orbital energy, but the total energy for the N-electron system is not the simple summation of these orbital energies.

Self Consistent Field Method

- Hartree-Fock (HF) uses orbital approximation of true wavefunction (cheap QM)
- Kohn-Sham density functional theory (KS-DFT) has same form as HF
- Self consistent field (SCF) used to determine orbital shapes
- As orbital shapes influenced by shape of all other orbitals, SCF method is a nonlinear process
- SCF must be solved iteratively, until P satisfy certain threshold.



Note: S is overlap matrix, C stands for Molecular coefficients matrix, P is density matrix, F is Fock operator, ϵ is one electron canonical orbital energy

Total energy

- ϵ_k at previous page is canonical orbital energy, but the total energy for the N-electron system is not the simple summation of these orbital energies. See below equation:

$$E_{HF} = \sum_{i=1}^N \hat{h}_{ii} + \sum_{i=1}^N \sum_{j=1}^{N/2} [2\hat{J}_{ij} - \hat{K}_{ij}] + V_{\text{nucl}}$$

- $\hat{h}_{ii}$ is one electron terms, which includes one electron kinetic energy and attraction energy to the nuclei; J and K are two-electron terms, which includes given electron and the rest of all electrons, J is Coulomb interactions, does not care about spin direction, K is exchange term, has to be all alpha or beta electrons with same direction; V is Coulomb interactions between nuclei.

Ground state and excited state

- Hartree Fock optimization will only give ground state energy
- If one wants to have excited states, initial occupied and virtual orbital swaps needs to happen, this will give you the initial excited state guess to begin, later, optimization mechanism, like maximum overlap method, sigma-SCF method, level shift techniques, etc, can be used to ensure optimization of this state not fall back down to the ground state.
- Variational principle is only true for ground state
- My understanding of using Slater determinants (SD) to find ground state is, you actually swap orbital orders and see which orbitals' order combination will give you the lowest energy, and canonical orbitals are arranged in Aufbau principle.
- The reason they use Slater determinants to represent the fermion wave function is because if you swap any molecular orbitals (two electrons, two columns in SD), it will change the sign, and it is antisymmetric properties, which required by Pauli exclusion principle in quantum system.

Post-HF methods

- HF is the simplest wave function method, normally is used to find ground state for the N-electron systems.
- Post-HF methods: perturbation theory, coupled-cluster method, configuration interaction method, non-orthogonal configuration interaction method, broken symmetry method, complete active space SCF method, etc.
- Post-HF methods use HF as reference method, then apply different strategies from there, and they have different computational cost and advantages; we use full CI as computational method as reference to evaluate all proposed method, but FCI itself can only be done for every small molecules, coupled-cluster is also treated as reference, it's less expensive than FCI, but still expensive in most of cases.